

Notes: Carrillo, Pomeranz, and Singhal (AEJ Applied, 2017)
Dodging the Taxman: Firm Misreporting and Limits to Tax Enforcement

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1 Big picture

1.1 One-sentence summary

This note summarizes Carrillo, Pomeranz, and Singhal (2017), which uses Ecuadorian corporate tax discrepancy notifications to show that third-party information can strongly increase reported revenue when taxpayers respond, but that weak credible enforcement and near one-for-one cost offsets sharply limit the effect on actual tax collection.

1.2 Research question

The paper asks whether third-party information is sufficient to improve tax enforcement in a low-capacity setting. More precisely, it studies three questions:

- How much firm misreporting is visible when the tax authority compares self-reported corporate income tax returns with third-party reports?
- Do firms amend their returns when the tax authority notifies them that revenue discrepancies have been detected?
- Conditional on amending, do firms actually increase taxable profits, or do they adjust other margins of the return to offset the revenue correction?

1.3 Core result

The central result is a two-part limitation of third-party enforcement:

1. **Enforcement limitation:** only a small share of notified firms file amendments, despite the fact that the tax authority has detected discrepancies.
2. **Multitasking limitation:** among firms that respond, reported revenue rises sharply, but reported costs rise almost one-for-one, so taxable profits and tax payments rise only modestly.

The paper therefore shows that third-party information is powerful but incomplete. Detection is not the same as enforcement, and monitoring one line item can shift evasion to less verifiable line items.

1.4 Why this paper matters

A large public finance literature emphasizes that modern governments collect high tax revenue because firms and employers act as fiscal intermediaries and generate third-party information. This paper shows that the usefulness of third-party information depends on the rest of the enforcement system. If the tax authority cannot credibly punish noncompliance, and if only revenue is cross-checked while costs remain hard to verify, then firms can “dodge” the enforcement intervention without paying much more tax.

2 Incremental contribution of the paper

2.1 Contribution relative to the tax evasion literature

The standard Allingham-Sandmo framework models evasion as a choice under audit risk. A recent literature emphasizes third-party information as a way to make evasion harder. Carrillo, Pomeranz, and Singhal add a key insight: third-party information has to be combined with credible

enforcement and coverage of multiple reporting margins. Otherwise, taxpayers can ignore the tax authority or shift evasion to unmonitored margins.

2.2 Contribution relative to third-party reporting papers

Many studies show that third-party information increases compliance when the tax authority can verify the tax base and impose credible penalties. This paper studies a weaker institutional environment, where detection does not automatically imply collection. Its incremental contribution is to separate three steps that are often bundled together:

- **Information production:** the tax authority observes a discrepancy between own reports and third-party reports.
- **Compliance response:** the firm decides whether to amend the return.
- **Net tax impact:** the firm may increase revenue but also increase costs, leaving taxable profit almost unchanged.

2.3 Contribution relative to multitasking and enforcement design

The paper provides a taxpayer-level example of a multitasking problem. If the authority verifies one margin, such as revenue, firms may adjust another margin, such as costs. This is closely related to Holmstrom and Milgrom’s logic: incentives and monitoring on one task can distort effort or manipulation on another task.

2.4 Contribution for development and fiscal capacity

For developing countries, the result is important because many governments invest heavily in digitization, electronic filing, and cross-checking systems. The paper shows that information technology is not a complete substitute for auditing, legal capacity, and credible punishment. The weakest link in the enforcement chain can dominate the final revenue effect.

3 Institutional setting and policy intervention

3.1 Corporate income taxation in Ecuador

The setting is Ecuador’s corporate income tax. Incorporated firms file annual corporate income tax returns and pay a flat tax rate on reported profits. They also file monthly VAT returns and annexes that report transactions with other firms. The tax authority, the Servicio de Rentas Internas (SRI), can use these annexes to construct third-party measures of a firm’s revenue and costs.

3.2 Third-party information

The SRI constructs third-party revenue using sources such as:

- purchases reported by client firms;
- credit card sales data;
- customs data;
- financial institution records.

Third-party cost information comes from imports and supplier reports. Importantly, these measures are incomplete. In particular, cash sales to final consumers are generally not third-party reported, and many cost items are difficult to verify without a traditional audit.

3.3 The notification policy

In 2011 and 2012, the SRI sent almost 8,000 discrepancy notifications to firms. These letters informed firms that the tax authority had detected revenue amounts attributable to the firm that exceeded the firm's self-reported revenue on previously filed corporate income tax returns. Firms were asked to file amended returns.

The intervention has three useful features:

- It refers to **previously filed returns**, so observed responses are reporting responses rather than real production responses.
- It introduces third-party enforcement in a large-scale systematic way.
- The 2008 round did **not** disclose the exact third-party amount, while the 2009 and 2010 rounds did, creating informative variation in the content of the notification.

3.4 Why the intervention is useful for identification

The authors compare pre-notification and post-notification filings. The identifying assumption is that firms would not have amended these old returns absent notification. Figure 1 supports this assumption: amendment rates are essentially zero before notification and among non-notified firms, but jump sharply for notified firms after the intervention.

4 Conceptual framework

4.1 Firm reporting variables

Let true revenue and costs be decomposed into third-party and self-reported components:

$$R = R_T + R_S, \quad C = C_T + C_S.$$

The firm reports revenue and costs:

$$\hat{R}, \quad \hat{C},$$

and reported profit is

$$\hat{\pi} = \hat{R} - \hat{C}.$$

True profit is

$$\pi = R - C.$$

The statutory tax rate is τ , so the firm pays $\tau\hat{\pi}$ if no evasion is detected.

4.2 Detection probability based on reported profit rate

The authors modify the standard evasion model by allowing the audit or detection probability to depend on the reported profit rate:

$$p\left(\frac{\hat{\pi} + \epsilon}{\hat{R}}\right), \quad p' < 0.$$

The intuition is that reporting very low profits relative to revenue is suspicious. A firm reporting \$100 of profit on \$1,000 of revenue looks more plausible than a firm reporting \$100 of profit on \$1,000,000 of revenue.

4.3 Why firms may underreport both revenue and costs

For a given desired level of reported profit $\hat{\pi}$, the firm wants to minimize the detection probability. Because the detection probability decreases when the reported profit rate rises, the firm prefers to report a smaller scale of activity:

$$\frac{\hat{\pi}}{\hat{R}} \text{ is higher when } \hat{R} \text{ is lower.}$$

Thus the firm has an incentive not only to underreport revenue but also to underreport legitimate costs. Underreporting costs helps make the firm appear smaller while preserving the desired level of reported profit.

4.4 Introducing third-party revenue reporting

Third-party revenue reporting creates a lower bound on reported revenue. If the firm reports $\hat{R} < R_T$, the tax authority can directly detect a discrepancy. In the model this produces a jump in the detection probability:

$$\hat{R} < R_T \Rightarrow p = \bar{p}.$$

Therefore, among firms that amend, the optimal response is to report

$$\hat{R} = R_T.$$

This predicts bunching of reported revenue at the third-party amount when the amount is disclosed.

4.5 Why cost offsets arise

Once the firm must set $\hat{R} = R_T$, it can still choose reported costs $\hat{C}$. Higher reported costs reduce taxable profit:

$$\hat{\pi} = R_T - \hat{C}.$$

But higher costs also reduce the reported profit rate and can increase audit risk:

$$\frac{R_T - \hat{C}}{R_T}.$$

The firm's choice of $\hat{C}$ balances lower tax liability against a higher probability of detection. If cost reports are hard to verify, firms can increase reported costs almost one-for-one with revenue corrections, sharply limiting the effect on tax collection.

4.6 Empirical predictions

The conceptual framework yields four empirical predictions:

1. In the full universe of firms, reported revenue should often be below third-party revenue.
2. Firms may also report costs below third-party cost measures because they want to underreport scale.
3. Among amending firms, disclosed third-party revenue amounts should generate bunching at R_T .
4. Revenue corrections should be accompanied by cost offsets when costs are less verifiable.

5 Data and empirical strategy

5.1 Administrative data

The authors combine several administrative sources:

- corporate income tax forms with self-reported revenue, costs, and tax liability;
- third-party reported revenue from client purchase reports, credit cards, customs, and financial institutions;
- third-party reported costs from imports and supplier reports;
- notification records, including which firms were selected and whether e-mails bounced.

The full sample contains economically active firms in 2008–2010, while the notification sample contains firms that received discrepancy notifications.

5.2 Main outcome variables

The key outcome variables are:

$$\begin{aligned}\Delta Revenue_i &= Revenue_{i,post} - Revenue_{i,pre}, \\ \Delta Cost_i &= Cost_{i,post} - Cost_{i,pre}, \\ \Delta Tax_i &= Tax_{i,post} - Tax_{i,pre}.\end{aligned}$$

The pre-period filing is the last filing before notification. The post-period filing includes amendments during the notification window.

5.3 Baseline causal comparison

The main causal design is a pre-post comparison for notified firms:

$$y_{it} = \alpha_i + \beta Post_t + \varepsilon_{it}.$$

Because the tax years were already closed and amendments were rare among non-notified firms, β is interpreted as the reporting effect of notification. Placebo tests on non-notified firms confirm that comparable amendments do not occur absent notification.

5.4 Revenue matching regression

For firms that adjust revenue, the authors estimate:

$$\Delta Revenue_i = \alpha + \beta \left(R_{T,i} - \hat{R}_{i,pre} \right) + \varepsilon_i.$$

A coefficient close to one means that firms increase reported revenue almost exactly by the amount of the detected discrepancy.

5.5 Cost matching regression

To measure offsets, the authors estimate:

$$\Delta Cost_i = \alpha + \gamma \Delta Revenue_i + \varepsilon_i.$$

A coefficient γ close to one means that firms raise reported costs almost dollar-for-dollar with the reported revenue increase. The implied increase in taxable profit per dollar of discrepancy is approximately

$$\beta(1 - \gamma).$$

Using the main estimates $\beta \approx 0.927$ and $\gamma \approx 0.962$, the implied taxable profit increase is only

$$0.927 \times (1 - 0.962) \approx 0.035,$$

roughly 3–4 cents per dollar of detected discrepancy.

6 Main empirical results

6.1 Widespread misreporting in the full sample

The full universe of firms shows substantial revenue underreporting relative to third-party information. In 2009–2010, 24 percent of filings report revenue below third-party amounts. The paper also documents direct evidence of cost underreporting: 26 percent of returns report costs below third-party amounts, and 5 percent of returns with positive tax liability do so.

6.2 Low amendment response

Only about 10–20 percent of notified firms filed an amended return. This is a striking result because the tax authority had already detected revenue discrepancies. The authors interpret the low response as evidence that enforcement capacity is limited even conditional on detection. In Ecuador, legal and resource constraints made it hard to prosecute non-compliers based only on e-mail notifications.

6.3 Strong revenue response among amenders

Among firms that amended and were told the exact third-party revenue amount, many reported exactly that amount. In the 2009 and 2010 rounds, 39 percent and 35 percent of amending firms exactly matched the third-party revenue number. Regression estimates show that adjusting firms increased reported revenue by about 92.7 cents per dollar of discrepancy.

6.4 Information content matters

The 2008 notification did not disclose the third-party revenue amount. In that round, firms adjusted revenue by only 36 cents per dollar of discrepancy on average, and only 6 percent matched the third-party amount exactly. This shows that firms respond not just to being contacted by the tax authority but to the specific information embedded in the notification.

6.5 Cost offsets almost eliminate the tax base increase

Firms offset the revenue adjustment by increasing reported costs. The main coefficient is 0.962: for each dollar of reported revenue increase, reported costs increase by 96.2 cents. This means that the notifications changed reported revenue but had only small effects on reported taxable profits.

6.6 Limited effect on tax collection

Amending firms increased reported revenue by \$86,036 on average and reported costs by \$80,184, leading to an average tax increase of only \$1,851. In the full notified sample, including non-amenders, the average tax increase was only \$293. The paper estimates that the notifications generated about \$2 million in total tax collection, compared with a potential of about \$193 million if all discrepancies had translated into taxable revenue without cost offsets.

7 Tables and figures from the paper

7.1 Table 1: Descriptive Statistics, All Firms, 2008–2010 and Table 2: Descriptive Statistics by Year, Notified and Amending Firms

Read:

- Table 1 describes the universe of economically active firms. The distribution is highly skewed: mean revenue is much larger than median revenue, and median tax liability is zero.
- Table 2 describes notified and amending firms. These firms have large negative revenue discrepancies: their self-reported revenue is below third-party revenue by economically meaningful amounts.
- The amendment rate is low: only 596, 249, and 420 firms amended in the 2008, 2009, and 2010 rounds, respectively.

Economic message: The treatment sample is not representative of all firms; it is enforcement-relevant because it contains firms with detectable revenue discrepancies. The low median tax liability and large discrepancies also explain why enforcement on this margin has potentially large fiscal stakes.

Revenue (\$000)	1,528 (41,512) [41]
Cost (\$000)	1,430 (40,913) [41]
Tax liability (\$000)	23 (617) [0]
For 2009–2010 only	
Reported revenue minus third-party revenue (\$000)	535 (13,490) [5]
Reported costs minus third-party costs (\$000)	677 (26,830) [24]
Percent of firms with positive third-party revenue	74
Percent of firms with positive third-party costs	90
Observations	199,043
Number of firms	88,505

Notes: Sample includes all economically active firms. Means are reported along with standard deviations in parentheses and medians in brackets. All monetary figures in thousands of US dollars. Information on third-party reports are for 2009–2010 only, the years for which we have complete information on third-party reporting.

Source: Authors' calculations

(a) Table 1: Descriptive statistics, all firms

	Fiscal year		
	2008	2009	2010
<i>Panel A. Notified firms</i>			
Revenue (\$000)	1,541 (6,118) [136]	624 (1,260) [121]	592 (1,237) [103]
Cost (\$000)	1,439 (5,398) [129]	583 (1,170) [113]	552 (1,153) [96]
Tax liability (\$000)	26 (171) [1]	10 (31) [1]	10 (28) [1]
Reported revenue minus third-party revenue (\$000)	−307 (1,741) [−42]	−176 (312) [−64]	−197 (388) [−66]
Observations	3,136	2,221	2,636
Number of firms with invalid e-mail addresses	—	159	163
<i>Panel B. Amending firms</i>			
Revenue (\$000)	1,402 (4,367) [179]	371 (944) [59]	418 (960) [34]
Costs (\$000)	1,331 (4,172) [163]	352 (887) [62]	398 (924) [33]
Tax liability (\$000)	18 (70) [1]	5 (17) [0]	5 (13) [0]
Reported revenue minus third-party revenue (\$000)	−217 (806) [−40]	−150 (299) [−54]	−173 (333) [−63]

(b) Table 2: Notified and amending firms

7.2 Figure 1: Amendment Rates and Figure 2: Revenue and Cost Discrepancies, All Firms with Positive Third-Party Information

Read:

- Figure 1 shows amendment rates around the notification date. Non-notified firms rarely amend, while notified firms exhibit a sharp post-notification increase.
- Figure 2 shows the distribution of differences between log self-reported and log third-party reported revenue and costs.
- The left mass below zero in revenue discrepancies indicates that many firms report revenue below third-party reports.
- The cost panel shows little sharp bunching and provides evidence that some firms underreport costs as well.

Economic message: Figure 1 validates the pre-post strategy, while Figure 2 shows why the enforcement intervention matters: there is substantial detectable misreporting, but not much pre-existing bunching at the third-party amount.

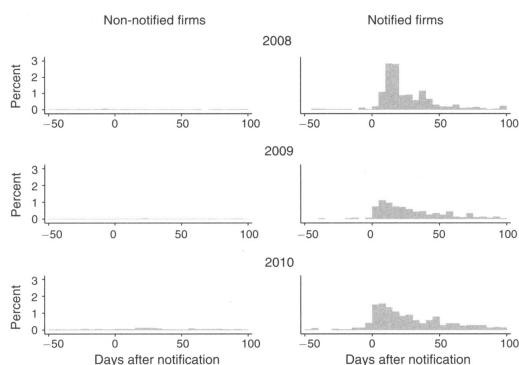


FIGURE 1. AMENDMENT RATES

Notes: The left column plots amendment rates for the universe of non-notified firms before and after the start of the intervention. The right column does the same for notified firms. Zero indicates the notification start date. For 2008, the start date is imputed (see footnote 19 for details).

(a) Figure 1: Amendment rates

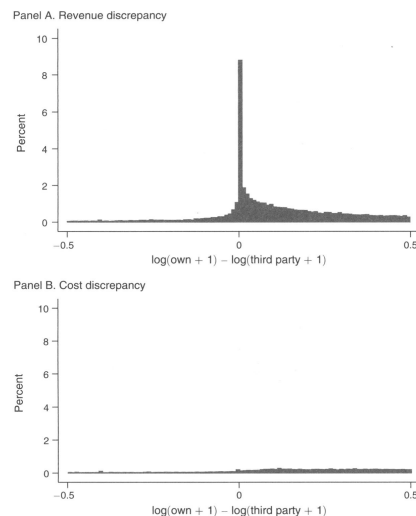


FIGURE 2. REVENUE AND COST DISCREPANCIES, ALL FIRMS WITH POSITIVE THIRD-PARTY INFORMATION

Notes: Both panels include data for 2009–2010, the years for which complete third-party information is available. Of the reports, 3.8 percent match revenue exactly and 0.1 percent match costs exactly. The sample in panel A includes the 60,067 active firms for which there are positive third-party revenues and the sample in panel B includes 74,614 active firms for which there are positive third-party costs. We restrict the range to -0.5 and 0.5 to focus on the range around zero.

(b) Figure 2: Revenue and cost discrepancies

7.3 Figure 3: Revenue Adjustment among Amending Firms and Figure 4: Revenue Matching

Read:

- Figure 3 shows strong bunching at zero in 2009 and 2010, when firms were given the SRI third-party revenue amount.
- Bunching is much weaker in 2008, when the exact third-party amount was not disclosed.
- Figure 4 shows that in 2009 and 2010, adjusting firms lie close to the 45-degree line: revenue increases closely match the discrepancy.

- The 2008 fitted line is much flatter, with a slope of 0.355, showing that disclosure of the amount changes behavior.

Economic message: Firms respond mechanically to the number disclosed by the tax authority. The specific information in the notification, not only the threat of scrutiny, drives the size of the revenue correction.

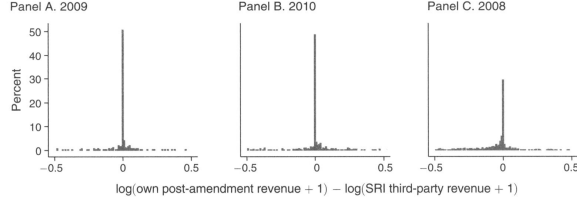


FIGURE 3. REVENUE ADJUSTMENT AMONG AMENDING FIRMS

Notes: Thirty-nine percent of firms match exactly for the 2009 round, 35 percent for the 2010 round, and 6 percent for the 2008 round. We add 1 in all log specifications to deal with zero self-reports. Bins are of size 0.01, and the top and bottom 1 percent of the sample are omitted when calculating bin heights for computational purposes.

(a) Figure 3: Revenue adjustment histograms

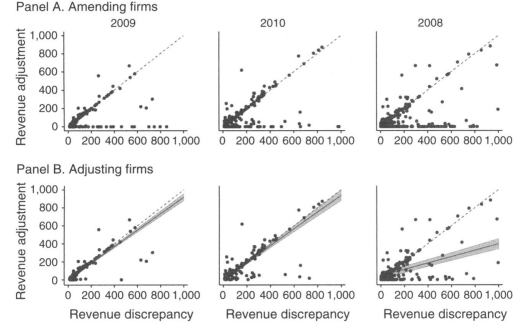


FIGURE 4. REVENUE MATCHING

Notes: The dashed line indicates a 45-degree line. In panel B, we also show a fitted line and a 95 percent confidence interval for the fitted line. Slopes are as follows: 0.916 for the 2009 round, 0.929 for the 2010 round, and 0.355 for the 2008 round. Axes are restricted to show zeros to one million, but the fitted line and confidence interval reflect the

(b) Figure 4: Revenue matching

7.4 Table 3: Treatment Effects and Robustness Tests for Adjusting Firms and Figure 5: Cost Matching

Read:

- Table 3, panel A reports a baseline revenue matching coefficient of 0.927 in 2009–2010.
- Table 3, panel B reports a baseline cost matching coefficient of 0.962 in 2008–2010.
- Figure 5 visually shows the cost offset: points lie nearly on the 45-degree line between revenue adjustments and cost adjustments.
- The cost matching result is extremely stable across robustness checks.

Economic message: The paper’s most important quantitative result is not that firms adjust revenue; it is that firms almost fully undo the tax-base effect by reporting higher costs. The gross compliance response is large, but the net tax-base response is small.

TABLE 3—TREATMENT EFFECTS AND ROBUSTNESS TESTS FOR ADJUSTING FIRMS:
REVENUE AND COST MATCHING

Specification	Coefficient	SE	Observations	R ²
<i>Panel A. Regressing revenue adjustment on revenue discrepancy (2009–2010)</i>				
Baseline (corresponds to Figure 4, panel B)	0.927	(0.105)	569	0.728
Robustness tests				
(1) Only firms notified in no more than one round	1.016	(0.093)	410	0.798
(2) Censoring to revenue discrepancies < \$1,000,000	0.900	(0.112)	555	0.623
(3) Censoring to revenue discrepancies < \$250,000	0.917	(0.052)	494	0.657
(4) Including negative revenue adjustments	0.926	(0.105)	579	0.285
(5) SRI letter measure of baseline $\bar{R}$	0.932	(0.098)	570	0.744
(6) Using only unadjusted variation in R_T	1.111	(0.151)	249	0.811
<i>Panel B. Regressing cost adjustment on revenue adjustment (2008–2010)</i>				
Baseline (corresponds to Figure 5)	0.962	(0.016)	978	0.982
Robustness tests				
(1) Only firms notified in no more than one round	0.960	(0.017)	737	0.983
(2) Censoring to revenue adjustments < \$1,000,000	0.975	(0.029)	959	0.882
(3) Censoring to revenue adjustments < \$250,000	0.897	(0.042)	885	0.677
(4) Including negative revenue adjustments	0.974	(0.015)	1,015	0.988
(5) SRI letter measure of baseline $\bar{R}$	0.942	(0.022)	1,053	0.962

Notes: Panel A shows linear regressions of revenue adjustment on revenue discrepancy for adjusting firms in 2009 and 2010. Panel B shows linear regressions of cost adjustment on revenue adjustment for adjusting firms in 2008–2010. A very small number of firms make negative revenue adjustments. The fourth robustness check shows what happens when they are included. We use the last filing prior to the notifications as the baseline measure of reported revenue. In some cases, the measure of baseline revenue provided to firms on the letters from the SRI differed slightly. The fifth robustness check uses the letter measure to calculate revenue adjustment. The measure of R_T provided to firms was hand-checked by SRI staff and adjusted in some cases. The sixth robustness check excludes such cases. All monetary figures are in US dollars. Standard errors clustered by firm are in parentheses, except for

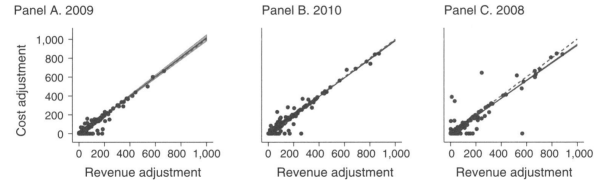


FIGURE 5. COST MATCHING

Notes: The dashed line indicates a 45-degree line. Also shown are a fitted line and a 95 percent confidence interval

(a) Table 3: Revenue and cost matching regressions

(b) Figure 5: Cost matching

7.5 Figure 6: Changes in Taxes among Adjusting Firms and Table 4: Treatment Effects on Overall Revenue, Cost, and Tax Liability, 2008–2010

Read:

- Figure 6 shows that changes in tax liability bunch around zero, despite large revenue and cost adjustments.
- Table 4 shows that amending firms increase revenue by \$86,036 and costs by \$80,184, producing only a \$1,851 tax increase.
- For the full notified sample, the average tax increase is only \$293 because most firms do not amend.

Economic message: The enforcement intervention changes reported line items but yields limited fiscal gains. The result is economically important because it separates accounting compliance from actual tax collection.

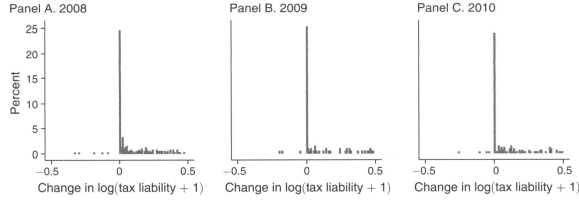


FIGURE 6. CHANGES IN TAXES AMONG ADJUSTING FIRMS

Notes: Histograms of the change in $\log(\text{tax liability} + 1)$ between prenotification and post-notification. We add 1 in

(a) Figure 6: Changes in taxes

TABLE 4—TREATMENT EFFECTS ON OVERALL REVENUE, COST, AND TAX LIABILITY, 2008–2010

	Revenue (1)	Cost (2)	Tax liability (3)
<i>Panel A. Amending firms</i>			
Post	86,036 (15,367)	80,184 (15,054)	1,851 (223)
Constant	872,195 (7,684)	828,644 (7,527)	11,254 (111)
Firm fixed effects	Yes	Yes	Yes
R^2	0.996	0.995	0.996
Observations	2,530	2,530	2,530
Number of firms	1,175	1,175	1,175
<i>Panel B. Notified firms</i>			
Post	13,616 (2,364)	12,690 (2,311)	293 (35)
Constant	973,408 (1,182)	908,571 (1,155)	16,226 (18)
Firm fixed effects	Yes	Yes	Yes
R^2	0.998	0.998	0.998
Observations	15,986	15,986	15,986
Number of firms	6,532	6,532	6,532

Notes: The baseline is the last filing before the notification and represents the amount after potential amendments during the post-notification window are taken into account. Dependent variables are in levels of US dollars. Standard errors clustered by firm are in parentheses.

Source: Authors' calculations

(b) Table 4: Overall treatment effects

7.6 Table 5: Placebo Tests, Simulated Treatment Effects for Non-notified Firms, 2008–2010

Read:

- Table 5 repeats the pre-post exercise for non-notified firms as if they had received notifications.
- There are no statistically significant changes in revenue, cost, or tax liability.
- Among non-notified firms that happened to amend, changes are imprecise and do not resemble the notified-firm pattern.

Economic message: The placebo evidence supports the causal interpretation: the observed revenue and cost adjustments are not part of a general trend in amended returns. They are specific responses to the SRI notifications.

TABLE 5—PLACEBO TESTS: SIMULATED TREATMENT EFFECTS FOR NON-NOTIFIED FIRMS, 2008–2010

	Revenue (1)	Cost (2)	Tax liability (3)
<i>Panel A. All non-notified firms</i>			
Post	2,565 (2,200)	−1,975 (3,292)	745 (820)
Constant	1,867,316 (1,100)	1,744,932 (1,646)	28,332 (410)
Firm fixed effects	Yes	Yes	Yes
R^2	0.889	0.875	0.863
Observations	293,370	293,370	293,370
Number of firms	66,872	66,872	66,872
<i>Panel B. Amending non-notified firms</i>			
Post	363,503 (376,976)	−279,836 (564,536)	105,568 (140,517)
Constant	11,996,503 (188,488)	11,771,793 (282,268)	49,020 (70,259)
Firm fixed effects	Yes	Yes	Yes
R^2	1.000	0.999	0.513
Observations	2,070	2,070	2,070
Number of firms	977	977	977

Notes: Linear regressions for the universe of all non-notified firms, supposing that they had been notified on the first day of the notification period for each year. Panel A includes all non-notified firms with prenotification filings and positive third-party revenue reports in the corresponding year. Panel B includes the firms from the previous panel that happened to amend during the post-notification window. All monetary figures in US dollars. Standard errors clustered by firm are in parentheses.

Source: Authors' calculations

Table 5: Placebo tests for non-notified firms

8 Interpretation of the evidence

8.1 Why the results are not simply honest corrections

The authors argue that the results are difficult to explain as honest errors or lazy reporting:

- A large share of notified firms initially reported zero revenue, even though discrepancies were often large.
- Revenue discrepancies frequently exceed tens of thousands of dollars.
- Revenue and cost adjustments are too tightly matched to look like accidental correction.
- There is little correlation between pre-notification profit rates and implied profit rates on the amended portion of the return.
- Firms tend to adjust harder-to-verify cost categories, such as administrative costs and “Other Administrative Costs.”

These patterns suggest strategic reporting rather than passive mistake correction.

8.2 Why enforcement capacity is central

Third-party information detects a discrepancy, but the tax authority still needs to induce compliance. In the Ecuadorian setting, the SRI sent e-mail notifications, and legal prosecution for failure to amend would have required more costly in-person notification procedures. Thus, the expected penalty for ignoring the e-mail could be low. The low amendment rate is therefore consistent with firms perceiving the threat as not fully credible.

8.3 Why partial third-party reporting creates cost-offset opportunities

The intervention targets revenue, but costs remain only partially verifiable. Suppose a firm receives a notification that its revenue should rise by ΔR . If the firm can report additional costs ΔC , then the increase in taxable profit is

$$\Delta\hat{\pi} = \Delta R - \Delta C.$$

If $\Delta C \approx \Delta R$, then $\Delta\hat{\pi} \approx 0$, and tax liability barely changes. This is exactly what the data show.

9 Policy implications

9.1 Third-party information is not a silver bullet

The paper’s main policy lesson is that third-party reporting is powerful only when embedded in a broader enforcement system. Information technology can identify discrepancies at low cost, but the government still needs legal and administrative capacity to force correction and to audit margins that are not third-party reported.

9.2 Tax-base design should depend on verifiability

If costs are easier to misreport than revenue, then profit taxation may perform poorly in low-capacity settings. A government may prefer tax bases that are easier to verify, even if they are less theoretically efficient. This logic connects the paper to Best et al. (2015), who show that production-inefficient turnover taxes can dominate profit taxes when cost evasion is severe.

9.3 The VAT self-enforcement mechanism may be weakened

A standard advantage of the VAT is that firms have incentives to report input purchases in order to claim credits. This paper’s evidence of cost underreporting shows that this self-enforcement mechanism can fail when firms have stronger incentives to understate overall scale or when cost deductions are not valuable because reported tax liability is already zero.

9.4 Complementarity between digital monitoring and traditional audits

Digital cross-checking and traditional audits are complements. Cross-checking identifies suspicious firms and suspicious margins; audits are needed to verify the margins that cross-checking cannot resolve, especially non-third-party costs. A low-capacity tax authority should therefore use third-party reports to target audit resources rather than expecting automated cross-checks to fully replace audits.

10 Relation to broader research themes

10.1 Information, enforcement, and organizational response

The paper is useful beyond tax because it studies what happens when an authority improves monitoring on one dimension. Agents do not passively comply; they re-optimize. This logic applies to many settings:

- firms respond to regulation by shifting activity to less monitored margins;
- banks respond to capital regulation by changing risk weights or off-balance-sheet exposure;

- organizations respond to performance metrics by gaming the measured variables;
- taxpayers respond to third-party information by moving evasion to unverifiable deductions.

10.2 Connection to multitasking

The key behavioral response is a multitasking response. The authority measures revenue more accurately; the firm increases cost misreporting. The general lesson is:

Monitoring one margin $\Rightarrow$ strategic substitution to other margins.

This is why the paper is relevant for research on governance, auditing, regulation, and incentive design.

10.3 Connection to fiscal capacity

The paper also provides a concrete mechanism for why fiscal capacity is hard to build. Capacity is not just about collecting information. It requires a chain:

information $\rightarrow$ credible threat $\rightarrow$ compliance $\rightarrow$ tax-base increase $\rightarrow$ revenue collection.

A failure at any link can sharply reduce final tax revenue.

11 Conclusion

11.1 Core takeaway

Carrillo, Pomeranz, and Singhal (2017) show that third-party information has large effects on reported revenue but small effects on tax collection when enforcement is weak and taxpayers can offset monitored margins with unmonitored margins. The paper therefore refines the optimistic view that third-party reporting automatically solves tax evasion.

11.2 What the paper teaches

The paper teaches four lessons:

1. Detection of misreporting is not the same as credible enforcement.
2. Gross reporting responses can be large even when net fiscal effects are small.
3. Monitoring one line item can induce strategic adjustment on another line item.
4. Digital tax administration needs to be paired with audit capacity and careful tax-base design.

11.3 Most important formula to remember

The core empirical accounting identity is:

$$\Delta TaxBase = \Delta Revenue - \Delta Cost.$$

The paper finds:

$$\Delta Revenue \approx 0.927 \times \text{discrepancy}, \quad \Delta Cost \approx 0.962 \times \Delta Revenue.$$

Thus:

$$\Delta TaxBase \approx 0.927(1 - 0.962) \approx 0.035,$$

meaning that only about 3–4 cents of taxable profit are created per dollar of detected revenue discrepancy among adjusting firms.

11.4 Final summary

The paper’s deeper message is that enforcement technologies change the game but do not end it. Firms learn what the tax authority can observe and then choose the least costly reporting response. Effective enforcement therefore requires not only better information but also credible punishment and coverage of the margins on which taxpayers can substitute evasion.